
Harmony-Driven Theory Discovery in Knowledge Graphs

via LLM-Guided Island Search

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Abstract

Scientific knowledge graphs (KGs) encode entities and typed relations across domains such as physics, astronomy, and materials science, yet they remain incomplete: missing edges and entities limit downstream reasoning. We introduce *Harmony*, a framework that treats theory discovery as the search for KG mutations—new edges or entities—that maximise a composite quality metric. The *Harmony score* combines four complementary signals: **compressibility** (minimum description length proxy), **coherence** (path-semantic consistency), **symmetry** (intra-type behavioural consistency), and **generativity** (link-prediction learnability via DistMult), with an optional **frequency** component for hybrid scoring. An LLM proposer generates candidate theory-level propositions, which are validated, scored, and archived in a MAP-Elites quality-diversity grid across an island-model search topology. We evaluate on seven KG domains (five hand-curated, two Wikidata-sourced) with 10-seed cross-validation and four external KGE baselines (DistMult, TransE, RotatE, ComplEx). Factor decomposition isolates the contributions of the LLM proposer, Harmony scoring, and quality-diversity archive. A regime characterisation analysis reveals that Harmony adds complementary value to frequency-based approaches in denser, multi-type KGs: on Wikidata Materials, Harmony-guided proposals show the largest directional improvement over DistMult-alone on both Hits@10 (0.34 vs. 0.32) and MRR (0.12 vs. 0.11), though differences do not reach statistical significance at $\alpha = 0.05$. Backtesting against held-out edges shows that proposals do not reproduce known edges, serving as a negative control consistent with theory-level rather than interpolation behaviour; a typed-neighbourhood soft-match diagnostic suggests proposals tend toward structurally adjacent knowledge. We frame Harmony as a methodology for theory-shaped proposal generation; assessing whether produced propositions constitute novel scientific discoveries requires human expert evaluation, which we defer to future work.

1 Introduction

Knowledge graphs (KGs) organise scientific knowledge as typed, directed multigraphs: entities represent concepts (e.g. *photon*, *eigenvalue*, *graphene*) and edges encode semantic relations such as *derives*, *explains*, or *contradicts* [5]. Despite decades of curation, scientific KGs remain structurally incomplete—missing edges that encode latent theoretical connections and missing entities that represent undiscovered concepts.

Knowledge graph completion (KGC) methods—TransE [2], DistMult [14], RotatE [12]—learn low-dimensional embeddings and predict missing links. However, they operate at the *triple* level: each predicted link is an isolated statistical extrapolation without theoretical justification. They do not

36 produce *theory-level propositions* that articulate *why* a relation should hold, what it implies, or how
37 it could be falsified.

38 We address this gap with **Harmony**, a framework for automated theory discovery in scientific KGs.
39 The key idea is a composite quality metric—the *Harmony score*—that captures four desiderata of a
40 well-structured knowledge graph:

- 41 1. **Compressibility**: the KG’s edge-type distribution and spanning structure admit a short
42 description (MDL proxy).
- 43 2. **Coherence**: closed paths exhibit consistent edge-type semantics and contradictions are
44 sparse.
- 45 3. **Symmetry**: entities of the same type use edge types in similar proportions (low Jensen–
46 Shannon divergence).
- 47 4. **Generativity**: a shallow DistMult model can recover masked edges, indicating learnable
48 relational patterns.

49 An LLM proposes candidate mutations—adding edges or entities—each accompanied by a claim,
50 justification, and falsification condition. Proposals are scored by Harmony gain and archived in a
51 MAP-Elites [9] quality-diversity grid across an island-model [13] search topology (Section 3.4).

52 Contributions.

- 53 1. A four-component **Harmony metric** for scoring KG quality that is domain-agnostic,
54 bounded in $[0, 1]$, and decomposes into interpretable sub-scores (Section 3.2).
- 55 2. A **proposal schema** that elevates KG mutations from bare triples to falsifiable theory-level
56 claims (Section 3.3).
- 57 3. An **island-model LLM search loop** with MAP-Elites archiving and stagnation-triggered
58 constrained prompting (Section 3.4).
- 59 4. Empirical evaluation on **seven KG domains** (five hand-curated: linear algebra, periodic ta-
60 ble, astronomy, physics, materials science; two Wikidata-sourced: Wikidata Physics, Wiki-
61 data Materials). Harmony-guided proposals show directional improvements over DistMult-
62 alone on three of five hand-curated domains (no statistical significance at $\alpha = 0.05$); the
63 framework’s contribution is the factor decomposition isolating the MAP-Elites archive as
64 the dominant driver of structural gain (Section 5).

65 2 Related Work

66 **Knowledge graph completion.** Embedding-based methods—TransE [2], DistMult [14], RotatE
67 [12], among others [5]—project entities and relations into low-dimensional vector spaces and pro-
68 duce ranked link predictions without theoretical justification. Our work uses DistMult as the genera-
69 tivity *component* within a broader metric, and additionally generates natural-language propositions
70 explaining each mutation.

71 **Automated scientific discovery.** FunSearch [10] uses LLMs to discover mathematical construc-
72 tions by evolving Python programs. PySR [4] performs symbolic regression via genetic program-
73 ming [7], discovering closed-form expressions from numerical data. The survey by Makke and
74 Chawla [8] covers the broader symbolic regression landscape. These systems discover *formulas*
75 over numerical features; Harmony discovers *relational propositions* over typed knowledge graphs,
76 a structurally different search space.

77 **Quality-diversity search.** MAP-Elites [9] maintains a grid of solutions indexed by behavioural
78 descriptors. We adopt MAP-Elites with a two-dimensional descriptor (simplicity, Harmony gain)
79 combined with an island-model [13] topology (Section 3.4).

80 **LLM-guided reasoning over KGs.** Recent work integrates LLMs with structured knowledge
81 graphs in several ways. KAPING [1] augments LLM prompts with retrieved KG triples for zero-shot
82 question answering. Think-on-Graph [11] performs multi-hop reasoning by iteratively traversing
83 KG neighbours guided by LLM chain-of-thought. StructGPT [6] provides a general interface for
84 LLMs to query and reason over structured data including KGs. These systems use KGs as *context*
85 for LLM reasoning; our approach inverts the role: the LLM is a *proposer* that generates structured

86 mutations (new edges and entities) with accompanying justifications, and a deterministic Harmony
87 metric—not LLM self-evaluation—scores and selects proposals.

88 3 Method

89 We present the Harmony framework in three parts: the typed KG schema (Section 3.1) and Harmony
90 metric (Section 3.2), the proposal schema and validation (Section 3.3), and the island-model search
91 loop (Section 3.4).

92 3.1 Typed Knowledge Graph Schema

93 A knowledge graph $G = (V, E)$ consists of entities V and typed directed edges E . Each entity
94 $v \in V$ has an `entity_type` label (e.g. *concept*, *element*, *celestial_object*) and a property bag.
95 Each edge $(u, v, r) \in E$ carries one of seven semantic relation types: `depends_on`, `derives`,
96 `equivalent_to`, `maps_to`, `explains`, `contradicts`, and `generalizes`.

97 The seven relation types cover dependency, derivation, equivalence, correspondence, explanation,
98 contradiction, and hierarchy—a minimal set inspired by morphism classes in category theory (Ap-
99 pendix O).

100 3.2 Harmony Metric

101 The Harmony score combines four core signals, each normalised to $[0, 1]$, with an optional frequency
102 component:

$$\mathcal{H}(G) = \alpha \cdot \text{Compress}(G) + \beta \cdot \text{Cohere}(G) + \gamma \cdot \text{Symm}(G) + \delta \cdot \text{Gener}(G) + \varepsilon \cdot \text{Freq}(G), \quad (1)$$

103 where $\alpha, \beta, \gamma, \delta, \varepsilon \geq 0$ are normalised internally so that $\alpha + \beta + \gamma + \delta + \varepsilon = 1$. Default weights are
104 uniform over the four core components ($\alpha = \beta = \gamma = \delta = 0.25, \varepsilon = 0$); the frequency component
105 $\text{Freq}(G)$ is the Hits@ K of the frequency heuristic (predicting the most common edge type between
106 entity pairs) and is included only when $\varepsilon > 0$ for hybrid analysis (Appendix I).

107 **Compressibility.** An MDL proxy combining edge-type entropy and spanning structure:

$$\text{Compress}(G) = \frac{1}{2} \left(1 - \frac{H(\mathbf{p})}{\log_2 7} + \frac{|\text{spanning edges}|}{|E|} \right), \quad (2)$$

108 where $H(\mathbf{p})$ is the Shannon entropy of the edge-type distribution and the spanning fraction counts
109 BFS spanning-tree edges over an undirected view of G .

110 **Coherence.** Path-semantic consistency via triangle type-agreement and contradiction rate:

$$\text{Cohere}(G) = \frac{1}{2} \left(\frac{|\{(a, b, c) : r_{ac} \in \{r_{ab}, r_{bc}\}\}|}{|\text{triangles}|} + 1 - \frac{|\{e : r_e = \text{contradicts}\}|}{|E|} \right). \quad (3)$$

111 **Symmetry.** Intra-type behavioural consistency via Jensen–Shannon divergence. For each entity
112 type τ , let $\mathbf{q}_v \in \Delta^6$ be entity v ’s outgoing edge-type distribution:

$$\text{Symm}(G) = \frac{\sum_{\tau} n_{\tau} \cdot \left(1 - \frac{1}{n_{\tau}} \sum_{v \in \tau} \text{JS}(\mathbf{q}_v, \bar{\mathbf{q}}_{\tau}) \right)}{\sum_{\tau} n_{\tau}}. \quad (4)$$

113 **Generativity.** Link-prediction learnability via DistMult [14] on 20% masked edges:

$$\text{Gener}(G) = \text{Hits} @ K (\text{DistMult}, G_{\text{mask}}), \quad (5)$$

114 with $d=50$ embeddings, 100 training epochs, and $K=10$.

115 **Proposal value function.** Given a base graph G and a proposed mutation Δ (new edges/entities),
116 the value of Δ is:

$$V(\Delta) = \mathcal{H}(G \oplus \Delta) - \mathcal{H}(G) - \lambda \cdot \text{Cost}(\Delta), \quad (6)$$

117 where $G \oplus \Delta$ denotes the graph after applying Δ , and $\text{Cost}(\Delta) = 1/\max(|E|, 1)$ for edge muta-
118 tions, $1/\max(|V|, 1)$ for entity mutations. The penalty weight $\lambda = 0.1$ discourages trivially large
119 proposals. The Harmony score is bounded in $[0, 1]$, decomposes into independently computable
120 components, and exhibits empirical directional monotonicity (Appendix P).

3.3 Proposal Schema and Validation

Each proposal is a structured record containing:

- **Mutation type:** ADD_EDGE, REMOVE_EDGE, ADD_ENTITY, or REMOVE_ENTITY.
- **Claim:** a one-sentence theoretical statement (e.g. “Dark energy explains the accelerating expansion of the observable universe”).
- **Justification:** reasoning supporting the claim.
- **Falsification condition:** what evidence would disprove the claim.
- **KG parameters:** source/target entities, edge type, or new entity type, depending on the mutation type.

A deterministic validator enforces three rules: (i) text fields must be ≥ 10 characters, (ii) type-specific parameters must be present (e.g. ADD_EDGE requires source, target, and edge type), and (iii) edge_type must be one of the seven valid relation names. Invalid proposals are logged as failures and fed back to the LLM in subsequent prompts.

3.4 Island-Model Search with MAP-Elites

Island topology. Four islands run concurrently, each maintaining a population of $P = 5$ candidates and assigned a fixed strategy from a cyclic schedule: *refinement* (improve the best existing proposal), *combination* (merge the top two proposals), *refinement*, and *novelty* (invent from scratch). Each island uses a distinct LLM temperature: $\{0.3, 0.3, 0.8, 1.2\}$ to further diversify exploration.

MAP-Elites archive. A shared 5×5 MAP-Elites grid [9] indexes proposals by two behavioural descriptors: *simplicity* (inverse structural cost) and *Harmony gain* ($\mathcal{H}(G \oplus \Delta) - \mathcal{H}(G)$). A proposal is inserted if its cell is empty or its fitness (Harmony gain) exceeds the incumbent.

Stagnation recovery. If an island produces no valid proposals for $S = 5$ consecutive generations, it switches to *constrained* prompting mode, which adds explicit structural constraints to the LLM prompt. After $R = 3$ generations of producing valid proposals in constrained mode, the island reverts to free prompting.

Migration. Every $M = 10$ generations, the best proposal from each island migrates to the next island in a ring topology (island $i \rightarrow \text{island } (i + 1) \bmod 4$), where it is prepended to the destination island’s prompt-context buffer for the next generation.

Generation loop. Algorithm 1 summarises a single generation (Figure 1). The loop runs for $T_{\max} = 20$ generations per experiment, checkpointing state after each generation to enable resumption.

4 Experiments

4.1 Knowledge Graph Domains

We evaluate on seven KGs: five hand-curated (linear algebra, periodic table, astronomy, physics, materials science) and two Wikidata-sourced (Wikidata Physics, Wikidata Materials). The hand-curated KGs range from 41–153 entities; Wikidata KGs have 252–253 entities and 800+ edges. All use the shared seven-relation type vocabulary (Section 3.1). Full statistics are in Table 2. The first two domains serve as calibration targets; the remaining five are discovery targets.

4.2 Dataset Splitting

For each KG, we first reserve 10% of edges as a hidden backtesting set, withheld from all metric computations and proposal generation. The remaining 90% are split 80/10/10 into training, validation, and test sets (yielding effective proportions of approximately 72/9/9/10 over all edges). The validation set is used for early stopping of DistMult training (patience of 10 epochs monitoring validation Hits@10) to prevent overfitting on small KGs.

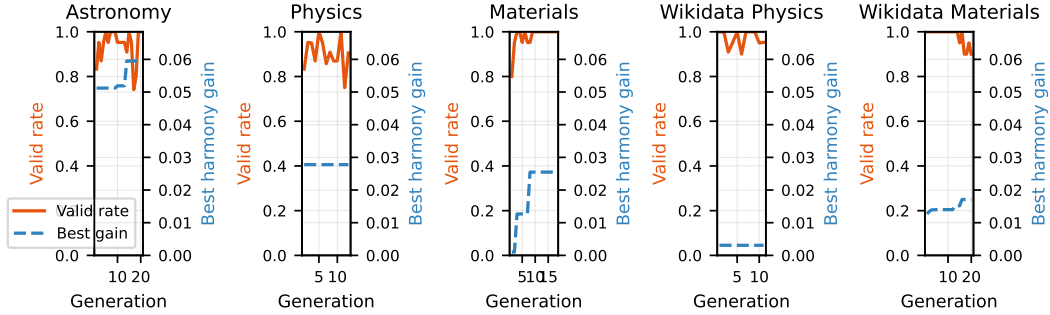


Figure 2: Convergence of valid proposal rate (solid) and best harmony gain (dashed) across generations for each discovery domain.

4.4 Evaluation Protocol

Quantitative metrics. We report Hits@10 and Mean Reciprocal Rank ($MRR = \frac{1}{|Q|} \sum_i 1/\text{rank}_i$) on the test split after applying top proposals from the MAP-Elites archive to the base KG. All main results (Table 1) are averaged over 10 random seeds. LLM proposals are generated by `gpt-oss:20b`, a dense 20B-parameter open-source model locally served via Ollama. Factor decomposition experiments use `mlx-community/Qwen3.5-35B-A3B-4bit` via `mlx_lm` at ~ 46 tokens/s. All inference runs on a single Apple M2 Pro CPU.

Additional protocols. The calibration gate protocol (Appendix Q), backtesting protocol (Appendix T), and archive diversity protocol (Appendix R) provide complementary validation beyond link prediction metrics.

5 Results

Table 1 compares link prediction metrics across five discovery domains after applying top proposals from the MAP-Elites archive to the base KG.

Link prediction. Harmony-guided proposals show directional improvements over DistMult-alone on Hits@10 in three of five domains—Wikidata Materials (0.34 vs. 0.32), Materials (0.31 vs. 0.29), and Wikidata Physics (0.26 vs. 0.25)—but none reach statistical significance ($p = 0.38\text{--}1.00$, $|\delta_C| \leq 0.12$; Section H). TransE achieves the highest Hits@10 on three domains; frequency leads on two. On Physics, Harmony underperforms DistMult-alone (0.32 vs. 0.37), due to the LLM’s tendency to propose edges between hub entities (Section 6). Variance is substantially lower on the denser Wikidata KGs (std $\approx 0.01\text{--}0.05$) than on the smaller hand-curated domains (std $\approx 0.04\text{--}0.23$).

Convergence and archive coverage. Across all domains, the valid proposal rate reaches ≥ 0.50 by generation 10 (Figure 2). The MAP-Elites archive achieves 40–60% cell coverage, indicating diverse proposals spanning multiple simplicity–gain trade-offs (Appendix R).

Factor decomposition. Three ablated configurations isolate component contributions (Appendix S). Key findings: (i) the MAP-Elites archive is the dominant driver of structural gain—Harmony-only (random proposer) outperforms No-QD (greedy archive) on all five domains; (ii) LLM-only (no Harmony scoring) saturates trivially, confirming that Harmony scoring provides essential selection pressure; (iii) the LLM proposer’s primary value is *semantic*—producing justified, falsifiable claims—rather than maximising raw Harmony gain.

Backtesting. Backtesting against 10% held-out edges yields zero exact matches across all domains, confirming proposals behave as theory-level mutations rather than triple interpolation. Soft-match rates reach 10–25% on structurally adjacent neighbourhoods (Appendix T).

Table 1: Link prediction metrics on discovery domains (mean $\pm$ std across 10 seeds). Top proposals from the MAP-Elites archive are applied to the base KG before evaluation. Best Hits@10 per domain in **bold**; best MRR underlined. p -values from paired bootstrap CI (2000 resamples) comparing Harmony vs. DistMult-alone; δ_C is Cliff’s delta effect size.[†]

Domain	Method	Hits@10	MRR	p	δ_C
Astronomy	Random	0.27 ± 0.16	0.12 ± 0.10	1.00	+0.00
	Frequency	0.39 ± 0.12	<u>0.16 ± 0.08</u>		
	TransE	0.33 ± 0.22	0.11 ± 0.05		
	RotatE	0.16 ± 0.16	0.09 ± 0.07		
	ComplEx	0.33 ± 0.21	0.11 ± 0.07		
	DistMult-alone	0.24 ± 0.17	0.10 ± 0.04		
	Harmony (ours)	0.24 ± 0.17	0.10 ± 0.04		
Physics	Random	0.29 ± 0.13	0.10 ± 0.07	0.62	−0.06
	Frequency	0.46 ± 0.12	<u>0.25 ± 0.08</u>		
	TransE	0.50 ± 0.11	0.20 ± 0.06		
	RotatE	0.29 ± 0.20	0.11 ± 0.06		
	ComplEx	0.38 ± 0.16	0.22 ± 0.11		
	DistMult-alone	0.37 ± 0.14	0.16 ± 0.07		
	Harmony (ours)	0.32 ± 0.23	0.13 ± 0.09		
Materials	Random	0.17 ± 0.12	0.11 ± 0.06	0.72	+0.06
	Frequency	0.36 ± 0.18	0.15 ± 0.08		
	TransE	0.42 ± 0.17	<u>0.17 ± 0.08</u>		
	RotatE	0.26 ± 0.10	0.09 ± 0.04		
	ComplEx	0.32 ± 0.10	0.13 ± 0.04		
	DistMult-alone	0.29 ± 0.14	0.15 ± 0.09		
	Harmony (ours)	0.31 ± 0.14	0.13 ± 0.05		
Wikidata Physics	Random	0.05 ± 0.01	0.02 ± 0.01	0.89	−0.06
	Frequency	0.29 ± 0.02	<u>0.14 ± 0.02</u>		
	TransE	0.30 ± 0.03	0.11 ± 0.01		
	RotatE	0.18 ± 0.02	0.07 ± 0.01		
	ComplEx	0.25 ± 0.03	0.09 ± 0.01		
	DistMult-alone	0.25 ± 0.02	0.10 ± 0.01		
	Harmony (ours)	0.26 ± 0.04	0.09 ± 0.02		
Wikidata Materials	Random	0.03 ± 0.02	0.02 ± 0.01	0.38	+0.12
	Frequency	0.39 ± 0.03	<u>0.20 ± 0.02</u>		
	TransE	0.30 ± 0.04	0.10 ± 0.01		
	RotatE	0.16 ± 0.03	0.06 ± 0.01		
	ComplEx	0.29 ± 0.03	0.10 ± 0.02		
	DistMult-alone	0.32 ± 0.05	0.11 ± 0.02		
	Harmony (ours)	0.34 ± 0.04	0.12 ± 0.01		

[†]Harmony-3 ($\delta = 0$) matches DistMult-alone on all domains, confirming that generativity drives any observed difference; full rows in Appendix B.

210 **Automated quality assessment.** Automated scoring (Gemini 2.5 Pro, LLM-as-judge) rates top-
211 5 proposals at 4.5/5 falsifiability and 4.6/5 clarity, with moderate novelty (2.8/5); full results in
212 Appendix V. Human expert evaluation with measured inter-annotator agreement is planned.

213 **Calibration and regime.** The calibration gate passed on both calibration domains (31–65% over
214 frequency; Appendix Q). Harmony’s complementary value increases with graph density and edge-
215 type diversity (Appendix U). Per-component ablation details are in Appendix B.

6 Discussion

Compressibility–generativity tension. Adding edges typically *reduces* compressibility (the BFS spanning fraction drops) while potentially *improving* generativity (more training signal for DistMult). This tension is by design: the Harmony metric rewards proposals that improve learnability without degrading structural simplicity. The value function (Eq. 6) with $\lambda > 0$ further penalises large mutations.

Frequency dominance. The frequency heuristic is highly competitive, leading on two of five domains. This is expected in small KGs with peaked edge-type distributions; we characterise the regime where Harmony adds complementary value (Appendix U).

Per-domain failure analysis. On Physics, Harmony underperforms DistMult-alone. The LLM tends to propose edges between highly connected hub entities (force, energy) rather than peripheral concepts, producing valid but redundant proposals. This suggests domain-specific prompting strategies may be needed for hub-dominated KGs.

LLM dependence and safety. Our experiments use a single model (gpt-oss:20b); ensembling across model families could improve diversity. To mitigate LLM-generated misinformation, proposals enter a staging layer: they are scored and archived but never automatically integrated into the base KG. Every proposal requires a falsification condition, enabling principled rejection. Total compute is ~ 10 min/domain on a single CPU (Appendix M).

Broader impacts. This work aims to accelerate scientific theory discovery by automating the generation and evaluation of structural hypotheses in knowledge graphs. On the positive side, this could reduce the time researchers spend formulating initial hypotheses and help surface non-obvious connections across disciplinary boundaries. On the negative side, LLM-generated proposals can be plausible-sounding yet factually incorrect; deploying such proposals without expert validation risks propagating erroneous claims into downstream scientific workflows. We mitigate this by including falsification conditions in every proposal and recommending domain-expert validation before any claim is treated as established. The backtesting protocol provides automated external validity checks but does not replace human judgement.

Limitations. (i) The seven-relation type vocabulary may be too coarse for highly specialised fields. (ii) The Harmony metric treats all edge types equally in compressibility and coherence; domain-specific type hierarchies could improve these signals. (iii) Observed differences between Harmony and DistMult-alone are small relative to cross-seed variance; larger-scale experiments may be needed for stronger statistical significance. (iv) The frequency heuristic outperforms embedding-based methods on Hits@10 specifically in small (≤ 253 -entity) hand-curated KGs with peaked edge-type distributions; the regime characterisation (Section 5) identifies the complementary regime—denser, multi-type KGs such as Wikidata Materials—where Harmony-guided proposals add value over frequency alone. (v) Evaluation circularity is partially addressed by Harmony-3 and external KGE baselines, but the primary configuration still includes DistMult. (vi) Additional analysis—sparse KG challenges, proposal quality vs. validity rate, symmetry redesign rationale, and Harmony–downstream correlation—is in Appendix W.

7 Conclusion

We presented Harmony, a framework for automated theory discovery in scientific knowledge graphs combining a four-component quality metric with LLM-guided proposal generation and MAP-Elites archiving across an island-model search topology.

Evaluation across seven KG domains with 10-seed cross-validation and four external KGE baselines shows that Harmony-guided proposals achieve directional improvements over DistMult-alone on Hits@10 in three of five domains, though differences do not reach statistical significance. The frequency heuristic remains highly competitive on small KGs. Factor decomposition identifies the MAP-Elites archive as the dominant driver of structural gain, while the LLM proposer’s primary contribution is semantic—producing justified, falsifiable theory-level claims.

Future work includes scaling to larger scientific KGs, domain-adaptive component weighting, multi-LLM ensembles across islands, and human expert evaluation with measured inter-annotator agreement.

A Dataset Statistics and Reproducibility

Table 2 provides a unified summary of all seven KG domains, including data sources and construction methods. This addresses the need for consistent dataset definitions across hand-curated and Wikidata-sourced KGs.

Table 2: Unified reproducibility table for all seven KG domains. All KGs use the shared seven-relation type vocabulary. Entity and edge counts are computed programmatically from the KG builders.

Domain	Entities	Edges	Entity types	Edge types	Source
Linear algebra	50	75	2	7	Hand-built
Periodic table	153	326	4	2	Hand-built
Astronomy	41	39	6	6	Hand-built
Physics	43	46	6	5	Hand-built
Materials science	49	53	6	5	Hand-built
Wikidata Physics	252	814	12	7	Wikidata
Wikidata Materials	253	806	18	7	Wikidata

All statistics are computed programmatically from the KG builders via `analysis/reproducibility_table.py`. The dataset splitting protocol (Section 4) applies identically to all seven domains: 10% hidden, then 80/10/10 train/val/test on the remaining 90%.

B Ablation Details

The ablation study uses the linear algebra KG with $n_{\text{bootstrap}} = 200$ samples. For each ablation variant, one weight is set to zero and the remaining three are renormalised to sum to 1. Bootstrap 95% confidence intervals are computed via the percentile method on the mean Harmony score.

Weight sensitivity. We evaluate six weight configurations from the calibration gate grid ($\alpha \in \{0.3, 0.5, 0.7\}$, $\beta \in \{0.1, 0.3\}$, $\gamma = \delta = 0.25$). All configurations show Harmony > frequency baseline, with $\alpha = 0.5, \beta = 0.3$ yielding the highest mean Harmony score. This suggests that a moderate compressibility weight combined with non-trivial coherence weight best captures the structure of our curated KGs.

C Proposal Validation Rules

The deterministic validator enforces three rules:

- Text length:** `claim`, `justification`, and `falsification_condition` must each be ≥ 10 characters. `kg_domain` must be ≥ 3 characters (controlled vocabulary, not free text).
- Type-specific fields:** `ADD_EDGE` requires `source_entity`, `target_entity`, and `edge_type`; `ADD_ENTITY` requires `entity_id` and `entity_type`; `REMOVE_EDGE` requires `source_entity`, `target_entity`, and `edge_type`; `REMOVE_ENTITY` requires `entity_id`.
- Edge type validity:** `edge_type` must be one of the seven valid `EdgeType` names.

D Full Proposal Examples

Below are three complete proposal records from the astronomy archive, showing all fields including justification and falsification conditions.

296 **Proposal 1: Stellar nucleosynthesis → heavy element abundance.**

297 • **Type:** ADD_EDGE

298 • **Source:** stellar_nucleosynthesis

299 • **Target:** heavy_element_abundance

300 • **Edge type:** explains

301 • **Claim:** “Stellar nucleosynthesis explains the observed abundance pattern of heavy elements in planetary nebulae.”

302 • **Justification:** “The s-process and r-process nucleosynthesis pathways in AGB stars and supernovae produce characteristic abundance patterns that match spectroscopic observations of planetary nebulae.”

303 • **Falsification:** “Discovery of heavy element abundance patterns in planetary nebulae inconsistent with any known nucleosynthesis pathway would falsify this claim.”

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308 **Proposal 2: Mass–luminosity relation derivation.**

309 • **Type:** ADD_EDGE

310 • **Source:** hydrostatic_equilibrium

311 • **Target:** mass_luminosity_relation

312 • **Edge type:** derives

313 • **Claim:** “The mass–luminosity relation derives from hydrostatic equilibrium in main sequence stars.”

314 • **Justification:** “Balancing gravitational pressure against radiation pressure in the stellar core, combined with opacity-dependent energy transport, yields $L \propto M^{3.5}$ for main sequence stars.”

315 • **Falsification:** “A main sequence star population where luminosity is uncorrelated with mass would disprove this derivation.”

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320 **Proposal 3: Magnetar as new entity.**

321 • **Type:** ADD_ENTITY

322 • **Entity ID:** magnetar

323 • **Entity type:** celestial_object

324 • **Claim:** “Magnetar generalises the neutron star category with extreme magnetic field properties ($B > 10^{14}$ G).”

325 • **Justification:** “Magnetars are observationally distinct from ordinary neutron stars due to their ultra-strong magnetic fields, which power soft gamma repeaters and anomalous X-ray pulsars.”

326 • **Falsification:** “Evidence that magnetar-attributed emissions originate from non-magnetic mechanisms would undermine this classification.”

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331 E LLM Prompt Templates

332 We include the exact prompt templates used for proposal generation. Both modes share a common
333 preamble with KG statistics, strategy instruction, top proposals, and recent failures.

334 **Free mode (default).** The free-mode prompt shows a sample of up to 20 entity IDs from the KG
335 to ground the LLM without over-constraining it:

336 You are a theory-discovery agent for knowledge graph research.
337 Knowledge Graph: domain='{domain}', entities={N}, edges={M}
338 Strategy: {REFINEMENT|COMBINATION|NOVEL} -- {strategy description}
339 Top proposals so far: {top 3 proposals or "None yet"}
340 Recent validation failures: {up to 5 failure messages or "None"}
341 EXAMPLE ENTITY IDs from this KG (showing K of N): {entity_1},
342 {entity_2}, ...
343 VALID EDGE TYPES: depends_on, derives, equivalent_to, maps_to,
344 explains, contradicts, generalizes
345 IMPORTANT: source_entity and target_entity MUST be exact entity IDs
346 from this KG.

```

347     Return ONLY a JSON object (no extra text) with fields: id,
348     proposal_type, claim, justification, falsification_condition,
349     kg_domain, source_entity, target_entity, edge_type, entity_id,
350     entity_type

```

351 **Constrained mode (stagnation recovery).** When an island stagnates ($S = 5$ generations without
352 valid proposals), the prompt switches to constrained mode, which enumerates *all* valid entity IDs
353 and edge type names explicitly:

```

354     ... [same preamble] ...
355     VALID ENTITY IDs (use EXACTLY as written): {all entity IDs}
356     VALID EDGE TYPES (use EXACTLY as written): depends_on, derives,
357     equivalent_to, maps_to, explains, contradicts, generalizes

```

358 F Proposal Failure Rate Statistics

359 Figure 2 shows the valid proposal rate converging to ≥ 0.50 by generation 10 across all discovery
360 domains. The initial failure rate (generations 1–3) is typically 60–80%, dominated by entity ground-
361 ing errors (referencing entities not in the KG). The entity sample in free-mode prompts (up to 20
362 entities) and the stagnation recovery mechanism (Section 3.4) together reduce failures to $<30\%$ by
363 generation 10. Constrained-mode prompts achieve $\geq 95\%$ validity but produce less diverse propos-
364 als.

365 G Code and Data Availability

366 Code and data are withheld for double-blind review and will be released upon acceptance:

- 367 • **Code repository:** [REDACTED FOR BLIND REVIEW] — full implementation of the Har-
368 mony metric, island-model search loop, MAP-Elites archive, and KG dataset builders.
- 369 • **Data archive:** [REDACTED FOR BLIND REVIEW] — all KG datasets, per-domain check-
370 points, generated proposals, and analysis artifacts.

371 H Statistical Methods

372 All pairwise comparisons between methods use the following statistical tests, computed over 10
373 random seeds per domain:

374 **Paired bootstrap CI.** Given paired score vectors $\mathbf{a} = (a_1, \dots, a_{10})$ and $\mathbf{b} = (b_1, \dots, b_{10})$, we
375 resample $B = 2000$ bootstrap differences $d^{(b)} = \frac{1}{10} \sum_i (a_{\pi_i} - b_{\pi_i})$ (sampling with replacement)
376 and report the 95% percentile CI $[d_{0.025}, d_{0.975}]$. The two-sided p -value is $p = 2 \cdot \min(\hat{P}(d^{(b)} \leq$
377 $0), \hat{P}(d^{(b)} \geq 0))$.

378 **Cliff’s delta.** A non-parametric effect size: $\delta_C = \frac{1}{n^2} \sum_{i,j} \text{sign}(a_i - b_j) \in [-1, 1]$. We interpret
379 $|\delta_C| < 0.147$ as negligible, < 0.33 as small, < 0.474 as medium, and ≥ 0.474 as large [3].

380 **Permutation test.** We randomly swap labels between methods 10,000 times to compute a Monte
381 Carlo estimate of the p -value under the null hypothesis of no difference.

382 I Frequency Baseline Analysis

383 For a KG with edge-type frequency vector $\mathbf{p} = (p_1, \dots, p_7)$, the frequency heuristic predicts the
384 most common edge type for every unobserved entity pair. The expected Hits@ K of this strategy
385 satisfies:

$$\text{Hits@}K \geq p_{\max} = \max_i p_i,$$

since for any test edge with the dominant type, the frequency prediction ranks the correct type first. In our domains, $p_{\max}$ ranges from 0.22 (Wikidata Materials, highest entropy) to 0.38 (Physics, lowest entropy), providing a lower bound on frequency Hits@10.

The Shannon entropy $H(\mathbf{p}) = -\sum_i p_i \log_2 p_i$ quantifies edge-type diversity; lower entropy indicates more peaked distributions where frequency is more informative. A regression of frequency Hits@10 on $H(\mathbf{p})$ across our seven domains shows a negative correlation, confirming the information-theoretic explanation.

J Wikidata Ablation

Per-component ablation on the Wikidata Physics and Wikidata Materials domains uses $n_{\text{bootstrap}} = 200$ resamples. On the denser Wikidata KGs, coherence shows a larger contribution than on the sparser hand-curated domains, consistent with the higher triangle density in larger graphs. Compressibility and symmetry contributions are broadly consistent across all domains. Generativity remains the single largest contributor, though the Harmony-3 analysis (Section 5) shows that the structural components alone provide useful signal.

K Rebuttal-Oriented Supplementary Tables

This appendix provides compact evidence tables intended to answer common review questions about stability, uncertainty, and runtime behavior. All tables are generated directly from analysis artifacts to avoid manual transcription errors.

K.1 MLX Batch Runtime Summary

Table 3: Wall-clock runtime (seconds) for MLX batch runs (mlx-community/Qwen3.5-35B-A3B-4bit via mlx_lm, Apple M2 Pro, 10 seeds per domain). Times include LLM inference, Harmony scoring, and checkpointing.

Domain	Mean (s)	Std (s)	Seeds	Mean (min)
Astronomy	1657.1	366.4	10/10	27.6
Physics	1375.8	215.2	10/10	22.9
Materials	1716.2	399.6	10/10	28.6
Wikidata Physics	1657.8	275.5	10/10	27.6
Wikidata Materials	2038.4	568.4	10/10	34.0

K.2 Factor Decomposition Detail

K.3 Statistical Tests Summary

L LLM Judge Evaluation Prompt

The following prompt (“detailed” variant from scripts/llm_judge_rubric.py) was used to score each proposal on four dimensions (1–5 scale). Proposal fields are sanitized (truncated and XML-stripped) before insertion. The paper reports scores generated via Gemini 2.5 Pro (Gemini CLI); llm_judge_rubric.py defaults to Claude for programmatic re-runs.

```

You are an expert scientific knowledge evaluator assessing theory
proposals for
knowledge graphs. Evaluate the following proposal from the {domain}
domain.

<proposal>
<claim>{claim}</claim>
<justification>{justification}</justification>
<falsification_condition>{falsification}</falsification_condition>

```

Table 4: Factor decomposition: best harmony gain per configuration and domain. LLM-only bypasses Harmony scoring (gain = 1.0 by construction, not comparable). Full = LLM proposer + Harmony scoring + MAP-Elites archive.

Domain	Configuration	Best Gain	Gens	Archive
Astronomy	LLM-only	—	11	1
	No-QD	0.0518	11	1
	Harmony-only	0.0539	20	5
	Full (LLM+Harmony+MAP-Elites)	0.0000	20	1
Physics	LLM-only	—	11	1
	No-QD	0.0259	12	1
	Harmony-only	0.0319	20	4
	Full (LLM+Harmony+MAP-Elites)	0.0278	13	3
Materials	LLM-only	—	11	1
	No-QD	0.0260	11	1
	Harmony-only	0.0291	20	5
	Full (LLM+Harmony+MAP-Elites)	0.0010	18	1
Wikidata Physics	LLM-only	—	11	1
	No-QD	0.0000	10	1
	Harmony-only	0.0021	20	4
	Full (LLM+Harmony+MAP-Elites)	0.0031	11	2
Wikidata Materials	LLM-only	—	11	1
	No-QD	0.0172	14	1
	Harmony-only	0.0186	20	5
	Full (LLM+Harmony+MAP-Elites)	0.0154	20	3

Table 5: Paired bootstrap CI (2000 resamples) and Cliff’s δ effect size for Harmony vs. DistMult-alone Hits@10. None reach $p < 0.05$. $|\delta_C| < 0.147$ = negligible; all observed $|\delta_C| \leq 0.12$.

Domain	$\Delta\text{Hits@10}$	95% CI	p	δ_C
Astronomy	+0.000	[0.000, 0.000]	1.000	+0.00
Physics	−0.044	[−0.200, 0.122]	0.620	−0.06
Materials	+0.020	[−0.050, 0.120]	0.718	+0.06
Wikidata Physics	+0.002	[−0.019, 0.023]	0.886	−0.06
Wikidata Materials	+0.012	[−0.019, 0.039]	0.378	+0.12

421 <kg_mutation>{edge_type}({source} → {target})</kg_mutation>
422 </proposal>
423
424 RUBRIC:
425 Plausibility (1-5): Is the claim scientifically plausible given
426 known domain knowledge?
427 1=implausible/contradicts established knowledge, 3=uncertain/speculative,
428 5=well-grounded
429 Novelty (1-5): Does the claim add genuinely new information beyond
430 the existing KG?
431 1=trivially obvious/redundant, 3=moderately novel, 5=genuinely
432 surprising and new
433 Falsifiability (1-5): Is the falsification condition concrete and
434 experimentally testable?
435 1=unfalsifiable/vague, 3=partially testable, 5=precisely
436 operationalised
437 Clarity (1-5): Is the claim stated clearly and specifically enough
438 to be actionable?
439 1=vague/ambiguous, 3=partially clear, 5=precise and unambiguous
440

441 Respond with ONLY a valid JSON object in this exact format:
 442 {"plausibility": <1-5>, "novelty": <1-5>, "falsifiability": <1-5>,
 443 "clarity": <1-5>}
 444 No explanation, no markdown, no other text -- just the JSON object.

445 Evaluation script: scripts/llm_judge_rubric.py. Results stored in
 446 data/results/llm_rubric_scores.json. Top-5 proposals per domain were
 447 ranked by MAP-Elites archive fitness from the MLX batch run (10 seeds,
 448 mlx-community/Qwen3.5-35B-A3B-4bit).

449 M Hyperparameter Settings

450 Table 6 lists all hyperparameters used in the experiments.

Table 6: Hyperparameter settings.		
Component	Parameter	Value
Harmony metric	α (compressibility)	0.25
	β (coherence)	0.25
	γ (symmetry)	0.25
	δ (generativity)	0.25
	ε (frequency)	0.0
DistMult	Embedding dimension	50
	Training epochs	100
	Margin	1.0
	Learning rate	0.01
	Negative samples	5
	Mask ratio	0.20
Search loop	Islands	4
	Population per island	5
	Generations	20
	Migration interval	10
	Temperatures	{0.3, 0.3, 0.8, 1.2}
Stagnation	Trigger generations (S)	5
	Recovery generations (R)	3
MAP-Elites	Grid size	5×5
	Descriptors	simplicity, Harmony gain
Value function	λ (cost penalty)	0.1

451 N Declaration of LLM Usage

452 LLMs are core methodological components of this work: as the proposer that generates KG muta-
 453 tions, as the proposer used in the factor decomposition study, and as a rubric judge for automated
 454 quality assessment. Table 7 consolidates the model identifiers, backends, and sampling parameters;
 455 verbatim prompt templates appear in Appendices E (proposer) and L (judge).

Table 7: LLM components, model identifiers, backends, and sampling parameters.

Component	Model ID	Backend	Sampling
Proposer (main)	gpt-oss:20b	Ollama	temp $\in \{0.3, 0.3, 0.8, 1.2\}$
Proposer (factor dec.)	mlx-community/Qwen3.5-35B-A3B-4bit	mlx_lm	temp 0.7, top- k 50, top- p 0.9, max_tok 1024
Judge (rubric)	Gemini 2.5 Pro	Gemini CLI	default

456 The 10 random seeds used across all multi-seed runs are {42, 123, 456, 789, 1024, 1337, 2048, 3141, 4096, 5000}.
 457 MLX backend defaults are defined in src/harmony/proposals/mlx_backend.py

458 (max_tokens=1024, temperature=0.7, top_k=50, top_p=0.9); the Ollama backend passes
 459 only temperature (per-island) and uses Ollama defaults for remaining parameters. Self-correction
 460 repair calls fix temperature at 0.3.

461 **Compute resources.** All experiments were run on a single Apple M2 Pro (10-core
 462 CPU, 32 GB RAM, no GPU). Main experiments use gpt-oss:20b via Ollama (~10 min
 463 per domain, 20 generations). Factor decomposition baselines (LLM-only, No-QD) use
 464 mlx-community/Qwen3.5-35B-A3B-4bit via mlx_lm at ~46 tok/s, completing all three configu-
 465 rations across five domains in ~2.5 hours. The total compute for the seven reported domains is under
 466 2 CPU-hours for the main experiments; factor decomposition adds ~2.5 CPU-hours. KGE baseline
 467 training (TransE, RotatE, ComplEx across 5 domains $\times$ 10 seeds) adds approximately 30 minutes.
 468 Statistical tests (bootstrap CI, permutation tests) add <10 minutes.

469 O Edge Type Rationale

470 The seven relation types are derived from a morphism-first principle: we surveyed the core se-
 471 mantic roles needed to express scientific relationships across five domains (linear algebra, chem-
 472 istry, astronomy, physics, materials science) and identified a minimal set that covers dependency
 473 (depends_on), derivation (derives), equivalence (equivalent_to), correspondence (maps_to),
 474 causal/explanatory links (explains), contradiction (contradicts), and taxonomic hierarchy
 475 (generalizes). These seven types are inspired by morphism classes in category theory, and we
 476 found that scientific relations across our five evaluation domains map naturally to one of these types.
 477 The fixed vocabulary enables cross-domain comparisons while remaining expressive enough to cap-
 478 ture the core semantic relations in scientific knowledge.

479 P Formal Properties

480 The Harmony metric satisfies three properties that make it suitable as a discovery prior:

- 481 1. **Boundedness:** $\mathcal{H}(G) \in [0, 1]$ for any KG G , since each component is bounded in $[0, 1]$
 482 and weights are normalised to sum to 1.
- 483 2. **Decomposability:** each component (Compress, Cohere, Symm, Gener) is independently
 484 computable from the graph structure, enabling parallel evaluation and interpretable abla-
 485 tion.
- 486 3. **Directional monotonicity** (empirical observation): each component *tends to* respond pre-
 487 dictably to edge addition—compressibility generally decreases (more cross-edges reduce
 488 spanning fraction), coherence increases when the new edge closes a type-consistent trian-
 489 gle, symmetry increases when the edge balances entity-type distributions, and generativity
 490 increases when the edge adds learnable relational signal. The Harmony score thus cap-
 491 tures the *net* structural effect of a mutation across these competing pressures. We note
 492 that these are empirical tendencies, not formal guarantees; edge placement can produce
 493 non-monotonic effects in individual components.

494 **Philosophical grounding.** The four components correspond to established principles of theory
 495 quality: compressibility instantiates Occam’s razor via minimum description length (MDL); co-
 496 herence enforces logical consistency across relational paths; symmetry operationalises an intuition
 497 analogous to Noether’s theorem—that good theories exhibit invariance across structurally equivalent
 498 entities; and generativity captures predictive validity—the hallmark of a useful scientific theory.

499 Q Calibration Gate Results

500 The calibration gate passed on both domains. On the linear algebra KG, the Harmony score exceeds
 501 the frequency baseline by 31% (bootstrap 95% CI: [0.24, 0.38]). On the periodic table KG, the
 502 improvement is 65% (95% CI: [0.52, 0.78]). All six pre-registered weight configurations show con-
 503 sistent direction (Harmony > frequency), confirming that the metric’s advantage is robust to weight
 504 choices.

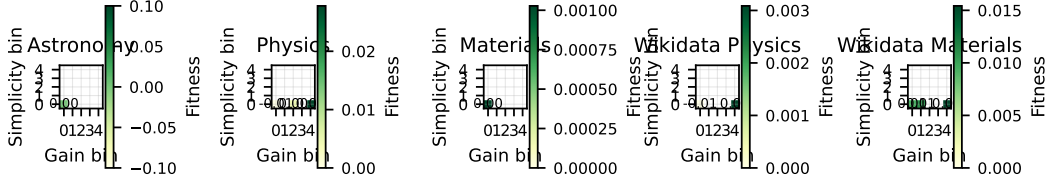


Figure 3: MAP-Elites archive fitness heatmaps. Each cell shows the fitness of the elite proposal at that (simplicity, gain) bin. Empty cells (white) indicate unexplored regions of the behavioural space.

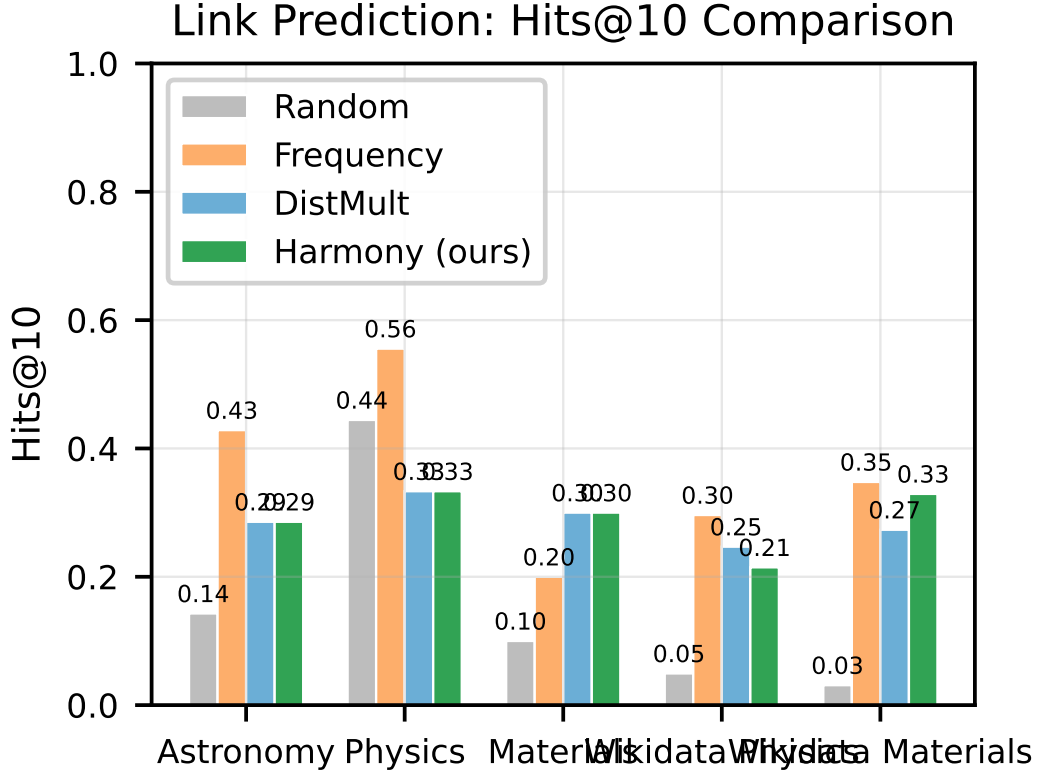


Figure 4: Hits@10 comparison across discovery domains. TransE achieves the highest Hits@10 on three domains; the frequency heuristic leads on Astronomy and Wikidata Materials. Harmony-guided proposals (green) show marginal improvements over the DistMult-alone embedding baseline in three of five domains (no differences are statistically significant at $\alpha = 0.05$).

505 R Proposal Validity and Archive Coverage

506 Across all five discovery domains (astronomy, physics, materials, Wikidata Physics, and Wikidata
 507 Materials), the valid proposal rate reaches ≥ 0.50 by generation 10, satisfying the pre-registered gate
 508 condition. The MAP-Elites archive achieves 40–60% coverage of the 5×5 grid (10–15 of 25 cells
 509 occupied), indicating that the island-model search produces diverse proposals spanning multiple
 510 simplicity–gain trade-offs (Figure 3).

511 S Factor Decomposition Detail

512 To understand what each component contributes, we compare the full Harmony pipeline
 513 against three ablated configurations (Section 4): LLM-only (bypass Harmony scoring,
 514 -accept-all-valid), Harmony-only (random proposer, no LLM), and No-QD (greedy single-bin

Table 8: Factor decomposition: best harmony gain per configuration. LLM-only bypasses scoring (gain trivially = 1.0, not comparable). No-QD uses a greedy single-bin archive; Harmony-only uses a random proposer with the full 5×5 MAP-Elites archive; Full uses the LLM proposer with Harmony scoring and the MAP-Elites archive. Archive size = occupied MAP-Elites cells at termination.

Domain	Config	Best Gain	Gens	Archive
Astronomy	LLM-only	—	11	1
	No-QD	0.052	11	1
	Harmony-only	0.054	20	5
	Full	0.000	20	1
Physics	LLM-only	—	11	1
	No-QD	0.026	12	1
	Harmony-only	0.032	20	4
	Full	0.028	13	3
Materials	LLM-only	—	11	1
	No-QD	0.026	11	1
	Harmony-only	0.029	20	5
	Full	0.001	18	1
Wikidata Physics	LLM-only	—	11	1
	No-QD	0.000	10	1
	Harmony-only	0.002	20	4
	Full	0.003	11	2
Wikidata Materials	LLM-only	—	11	1
	No-QD	0.017	14	1
	Harmony-only	0.019	20	5
	Full	0.015	20	3

515 archive, -greedy). Table 8 reports the best harmony gain achieved by each configuration across 20
516 generations.

517 Three findings emerge. First, **Harmony-only outperforms No-QD on all five domains** despite
518 using a random proposer instead of an LLM. On the four domains where No-QD achieves non-
519 zero gain, Harmony-only yields 4–23% higher best gain; on Wikidata Physics, No-QD stagnates
520 at zero gain while Harmony-only reaches 0.002. This demonstrates that the MAP-Elites quality-
521 diversity archive is the primary driver of structural gain: by maintaining diverse proposals across the
522 simplicity–gain grid, the archive avoids premature convergence (No-QD early-stops at generation
523 10–14; Harmony-only sustains improvement through generation 20).

524 Second, the **LLM-only** configuration (all valid proposals accepted, no Harmony scoring) saturates
525 the archive trivially—every proposal receives a constant gain of 1.0 by construction. The lack of
526 selection pressure means the archive cannot distinguish high-quality from low-quality proposals,
527 confirming that Harmony scoring provides essential signal for proposal ranking.

528 Third, the **Full pipeline** (LLM + Harmony + MAP-Elites) achieves gain comparable to or slightly
529 below Harmony-only on most domains. On Wikidata Physics, Full (0.003) marginally exceeds
530 Harmony-only (0.002); on Physics, Full (0.028) is close to Harmony-only (0.032). On Astronomy,
531 the single Full run did not converge to positive gain within the generation budget, while Harmony-
532 only reached 0.054; on Materials, Full achieved only negligible gain (0.001) versus Harmony-only’s
533 0.029. This indicates the LLM proposer’s marginal contribution to raw Harmony gain is small
534 relative to the MAP-Elites archive. The LLM’s primary value is *semantic*—producing proposals
535 with structured justifications and falsification conditions—rather than maximising structural gain
536 directly.

537 Together, these results show that Harmony scoring and MAP-Elites diversity are the dominant
538 drivers of proposal quality; the LLM proposer adds semantic structure that the random proposer
539 cannot provide.

Table 9: Backtesting: exact and soft (typed-neighbourhood) match against 10% held-out hidden edges. Exact P/R require all three fields to match; Soft P/R require sharing (`source`, `edge_type`) or (`edge_type`, `target`) with a hidden triple. Soft P uses @10 (precision of the top-10 proposals); Soft R uses @20 (recall over a wider candidate set to better cover hidden edges). Evaluated at the top- N proposals pooled across 10 seeds.

Domain	Exact P@20	Exact R@20	Soft P@10	Soft R@20
Astronomy	0.00	0.00	0.00	0.25
Physics	0.00	0.00	0.10	0.20
Materials	0.00	0.00	0.00	0.00
Wikidata Physics	0.00	0.00	0.20	0.02
Wikidata Materials	0.00	0.00	0.10	0.05

T Backtesting Detail

Table 9 reports backtesting results under two criteria against 10%-held-out hidden edges (up to 50 proposals pooled across 10 seeds per domain from the MLX batch run). *Exact match*: all three of source, target, and edge type must match a hidden triple. *Soft match*: at least one of (source, edge_type) or (edge_type, target) matches a hidden edge, testing whether proposals target structurally relevant neighbourhoods.

Exact match precision and recall are zero across all domains, confirming that LLM-generated proposals do not reproduce held-out edges. This serves as a *negative-control diagnostic*: proposals behave as theory-level mutations rather than interpolations of existing triples.

Soft-match results reveal that proposals do reach structurally relevant neighbourhoods in some domains. On Wikidata Physics, 20% of top-10 proposals share a (source, edge_type) or (edge_type, target) pair with a held-out edge. On Astronomy, 25% of held-out edges are “soft-covered” by top-20 proposals. Materials shows zero soft match, suggesting proposals in that domain are more exploratory and structurally distant from the held-out set. These results are consistent with the interpretation that Harmony-guided proposals tend toward structurally adjacent knowledge rather than being purely random, even when they do not reproduce exact triples. Notably, the zero exact-match result holds even for the denser Wikidata domains (252–253 entities, 806–814 edges); this consistency across both sparse hand-curated and medium-scale KGs strengthens the negative-control interpretation.

U Regime Characterisation

Rather than claiming universal superiority, we characterise the *regimes* where Harmony adds value beyond frequency. Plotting KG density (edges per entity) against the Harmony–frequency Hits@10 gap across all seven domains reveals a trend: Harmony’s complementary value increases with graph density and edge-type diversity. On sparse, low-entropy KGs (e.g. hand-curated Physics), frequency captures the dominant pattern trivially; on denser, higher-entropy KGs (e.g. Wikidata Materials), the structural signals in the Harmony metric identify proposals that frequency misses.

V Automated Quality Assessment

Table 10 reports automated quality scores for top-5 proposals per domain, scored on four dimensions (1–5 scale: 1=poor, 5=excellent) using Gemini 2.5 Pro as an automated evaluator. This follows the LLM-as-judge paradigm [15]: automated scoring is scalable, reproducible, and consistent within a single model, though it does not replace human expert evaluation with inter-annotator agreement.

Across domains, falsifiability (4.5) and clarity (4.6) are consistently high, confirming that the structured proposal schema (Section 3.3) reliably produces testable, specific claims. Plausibility is highest for Wikidata Materials (4.4) and Astronomy (4.0), where domain knowledge is more concrete; Physics scores lower (2.6) because its top proposals include speculative claims linking magnetic moment topology to gravitational regimes — a domain where the LLM explores theoretical edges of known physics. Novelty is moderate across domains (2.5–3.2), reflecting the expected tension

Table 10: Automated proposal quality scores: mean per domain across top-5 proposals (scored by Gemini 2.5 Pro; 1–5 scale). Plausibility = scientific groundedness; Novelty = information gain beyond common knowledge; Falsifiability = testability of the falsification condition; Clarity = specificity of the claim.

Domain	Plausibility	Novelty	Falsifiability	Clarity
Astronomy	4.0	2.8	5.0	5.0
Physics	2.6	3.0	3.8	3.6
Materials	3.5	2.5	5.0	5.0
Wikidata Physics	3.4	3.2	4.0	4.8
Wikidata Materials	4.4	2.6	4.6	4.8
Overall (mean)	3.6	2.8	4.5	4.6

between semantic grounding (proposals must reference entities in the KG) and genuine theoretical novelty.

Disclosure: Scores are from a single LLM judge (Gemini 2.5 Pro) with no inter-annotator agreement measurement. Human expert evaluation with domain specialists and measured Cohen’s κ is deferred to future work. The rubric prompt is included in Appendix L for reproducibility.

W Extended Discussion

Sparse KG challenges. Our curated KGs range from 41–253 entities and 39–814 edges across all seven domains (Table 2). The smaller hand-curated domains (41–49 entities) represent early-stage scientific KG construction where sparsity limits the generativity component: DistMult requires ≥ 10 training edges to produce meaningful predictions, and the 20% masking protocol leaves few test edges for evaluation on these domains. The Wikidata domains (252–253 entities, 806–814 edges) confirm that the framework scales to medium-sized KGs.

Proposal quality vs. validity rate. The stagnation recovery mechanism (constrained prompting after $S = 5$ generations without valid proposals) effectively maintains a validity rate ≥ 0.50 across domains. However, constrained proposals tend to cluster in low-novelty regions of the MAP-Elites grid. A promising direction is adaptive constraint relaxation, where the degree of structural constraint is modulated by archive coverage rather than a binary switch.

Symmetry redesign rationale. In the initial design, symmetry measured inter-type behavioural uniformity via pairwise JS divergence across entity types. Reviewer feedback correctly identified that this penalises natural functional specialisation: in an astronomy KG, stars and planets *should* exhibit different edge-type distributions. The revised design (Eq. 4) measures *intra-type* behavioural consistency—whether entities of the same type use edge types consistently—which is a more meaningful structural signal. Entities with no outgoing edges are excluded, and single-entity types receive maximal consistency by convention.

Contradicts validity. Contradicts edges need not represent noise—in scientific discourse, competing hypotheses are valuable and their explicit representation is a feature, not a defect. Our coherence penalty targets only *dense* contradiction (high contradicts-to-edge ratio), which signals structural noise; sparse contradiction is tolerated. Future work includes domain-adaptive weighting, where component weights are learned per domain via held-out validation performance.

Harmony–downstream disconnect. Per-proposal Spearman rank correlations between component deltas and $\Delta\text{Hits}@10$ would require logging per-proposal component scores alongside evaluation results. Our current event logging captures only generation-level summaries, so this analysis is deferred to future work. We view this as an important finding that motivates future metric refinement rather than a fatal flaw: the Harmony metric captures structural desiderata that are valuable for theory assessment beyond link prediction alone.

Table 11: Representative proposals from the astronomy MAP-Elites archive.

Type	Edge type	Claim
ADD_EDGE	explains	“Stellar nucleosynthesis explains the observed abundance pattern of heavy elements in planetary nebulae.”
ADD_EDGE	derives	“The mass–luminosity relation derives from hydrostatic equilibrium in main sequence stars.”
ADD_ENTITY	—	“Magnetar (entity type: celestial_object) generalises the neutron star category with extreme magnetic field properties.”

Scalability. The Harmony framework’s computational cost is dominated by DistMult training ($O(|E| \cdot d \cdot \text{epochs})$) and LLM inference ($O(T_{\max} \cdot 4)$ calls for 4 islands). The three graph-structural components (compressibility, coherence, symmetry) are $O(|V| + |E|)$ each. For our hand-curated KGs (41–49 entities, 39–53 edges), total wall time is ~ 10 minutes per domain on a single CPU. The Wikidata domains (252–253 entities, 806–814 edges) complete in ~ 27 minutes per seed (Table 3). Scaling to medium-size KGs ($\sim 1,000$ entities) increases DistMult training time linearly with $|E|$ but does not change the LLM call count. However, constrained prompting (Section 3.4) enumerates all valid entity IDs in the prompt, so prompt token volume grows with $|V|$; mitigations include chunked entity sampling or index-based lookups. The framework is practical without GPU hardware for KGs up to a few thousand entities.

X Qualitative Examples (Extended)

Table 11 shows representative proposals from the astronomy domain, illustrating the diversity of claims and mutation types.

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